Bank Term Deposit Subscription Prediction using CatBoost

1. Problem Description

Banks regularly run direct marketing campaigns to encourage customers to invest in term deposits. However, reaching out to every customer is both expensive and inefficient, as only a small percentage actually choose to subscribe. As a result, identifying customers who are more likely to respond positively has become essential for improving marketing effectiveness and reducing unnecessary costs.
This project focuses on developing a supervised machine learning model to predict whether a customer will subscribe to a bank term deposit using historical marketing data. By combining advanced machine learning methods with explainable AI techniques, the system not only improves prediction accuracy but also ensures that the results are transparent and easy to understand, supporting more informed decision-making.

2. Dataset Description

The dataset used in this project is the Bank Marketing Dataset sourced from the UCI Machine Learning Repository. It contains real-world data gathered from direct marketing campaigns carried out by a Portuguese banking institution, capturing customer characteristics and campaign details that influence term deposit subscription decisions.

Attribute	Description
Instances	45,211
Features	16
Target Variables	Y (Subscribed / Not Subscribed)
Feature Types	Categorical and Numerical
Ethical Consideration	Public, anonymized dataset

3. Methodology

I. Machine Learning Algorithm

This project uses CatBoost, a gradient boosting algorithm that is well suited for working with datasets containing many categorical features. Unlike traditional models such as logistic regression or decision trees, CatBoost handles categorical variables directly, eliminating the need for complex encoding methods. It applies an ordered boosting approach to reduce target leakage and improve the reliability of predictions. In addition, CatBoost helps control overfitting and requires very little feature preprocessing, making it an effective and practical choice for structured banking data.

II. Model Training

The dataset was split into training and testing sets, where 80% of the data was used to train the model and the remaining 20% was used for evaluation. Stratified sampling was employed to maintain the original class distribution across both datasets. The CatBoost model was then trained using its default parameters, with a focus on achieving reliable and effective classification performance.

III. Evaluation Metrics

Because the dataset is imbalanced, with relatively few customers choosing to subscribe to the term deposit, several evaluation metrics were used to properly assess the model's performance. Accuracy was used to measure overall prediction correctness, while the F1-score helped balance precision and recall. In addition, ROC-AUC was selected as the primary evaluation metric, as it offers a more reliable assessment of model performance when dealing with imbalanced classes.

4. Results

The trained CatBoost model achieved the following performance:

Metric	Value
Accuracy	0.91
F1-Score	0,56
ROC-AUC	0,93

5. Interpretation of Results

The model achieves a high accuracy of 91%, indicating strong overall prediction performance. However, the F1-score is comparatively lower because the dataset is imbalanced, with far fewer customers subscribing to the term deposit than those who do not. The ROC-AUC score of 0.93 shows that the model is highly effective at distinguishing between subscribers and non-subscribers, making it the most reliable metric for this task. Overall, the results demonstrate that the model successfully captures customer subscription behavior.

6. Explainability and Model Interpretation

To improve transparency, Explainable AI (XAI) techniques were incorporated using LIME (Local Interpretable Model-Agnostic Explanations). LIME helps explain individual predictions by approximating the complex CatBoost model with a simpler, interpretable model at a local level, making it easier to understand why a particular customer received a specific prediction. Additionally, global feature importance analysis revealed that call duration, contact method, and campaign timing had the greatest influence on the model's decisions. These results are consistent with real-world banking marketing insights and practical experience.

7. Critical Discussion

Although the model performs well overall, it does have some limitations. The imbalance in the dataset makes it more difficult for the model to accurately identify customers who choose to subscribe, which can reduce recall for this group. Additionally, the model is trained on historical campaign data, meaning it may not fully capture future changes in customer behavior or market conditions. There is also a risk of bias if predictions are relied upon without human judgement. For these reasons, ethical use of the model requires transparency, responsible implementation, and ongoing human oversight to ensure fair and informed decision-making.

8. Front-End System

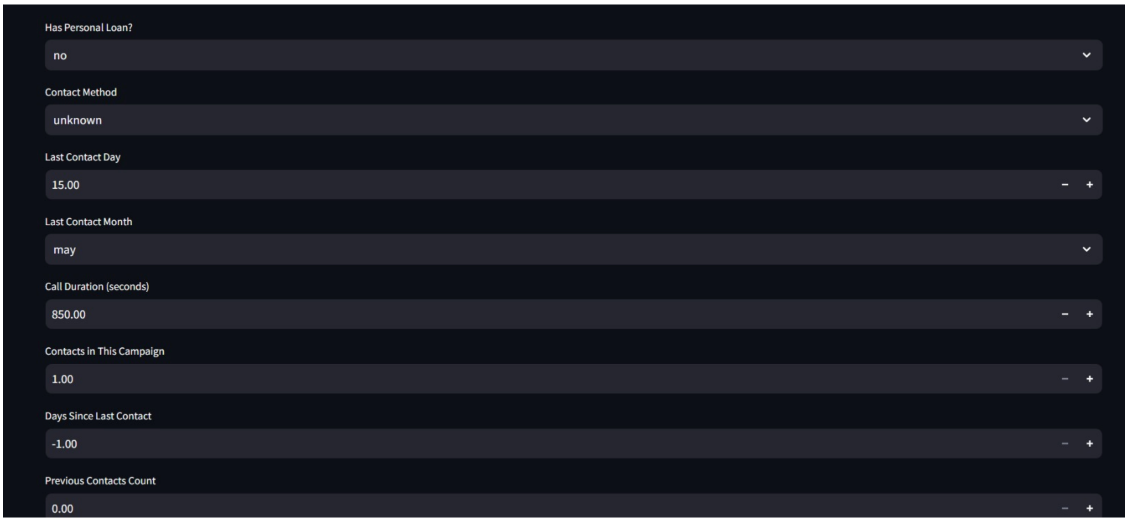
A Streamlit-based web application was created to ensure that the system can be easily used by non-technical users. Through this application, users can input customer details, view predictions on term deposit subscriptions, and explore clear explanations using interactive Plotly charts. This user-friendly design improves usability and enables real-time, well-informed decision-making.

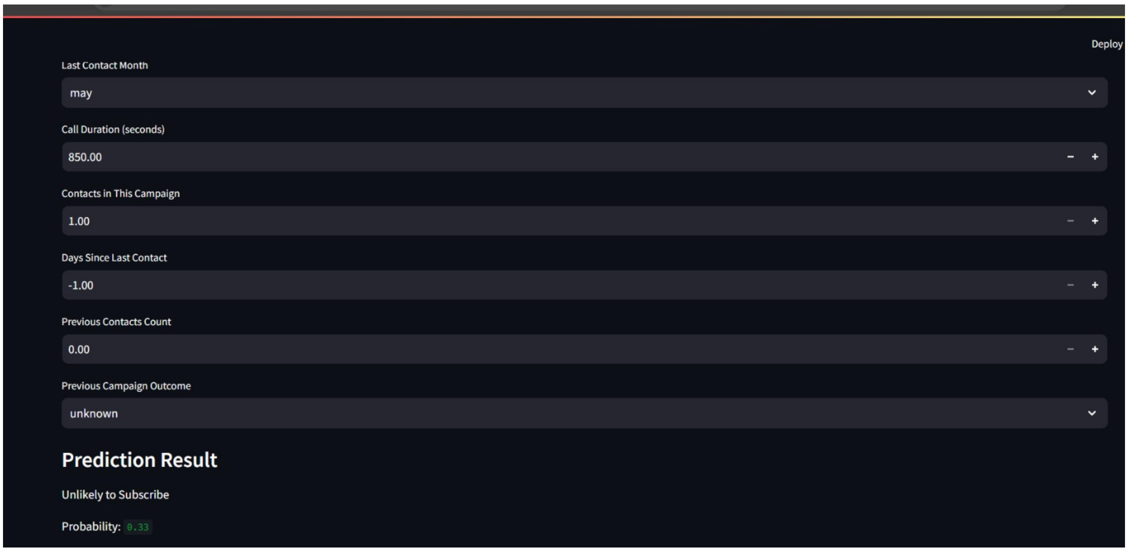
9. Conclusion

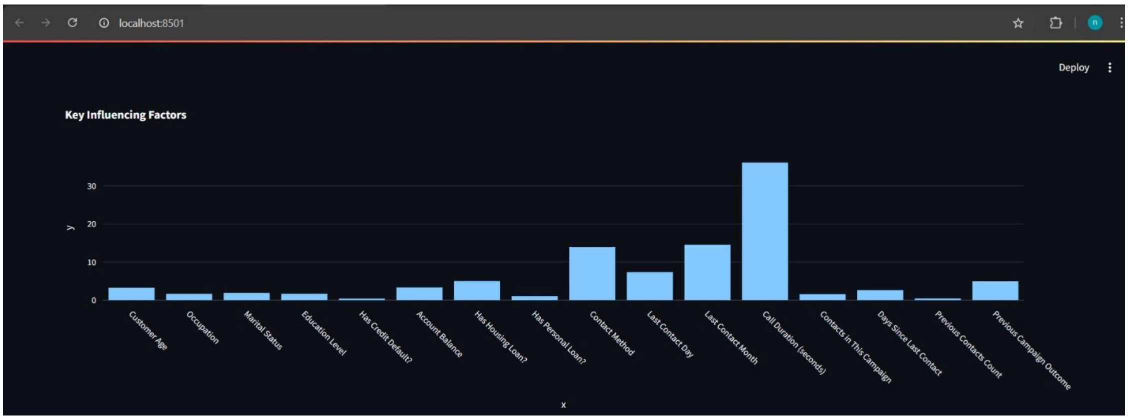
This project shows how combining CatBoost with Explainable AI techniques can effectively support bank marketing campaigns by improving both decision-making and operational efficiency. The developed model delivers strong predictive performance while successfully handling structured and imbalanced banking data. By using explainability tools such as LIME, the system provides clear insights into model predictions, helping to build trust and understanding. Additionally, the Streamlit-based interface makes the solution easy to use for non-technical users, enabling real-time predictions and interpretation. Overall, the project demonstrates the practical benefits of applying advanced machine learning with explainability in real-world banking.

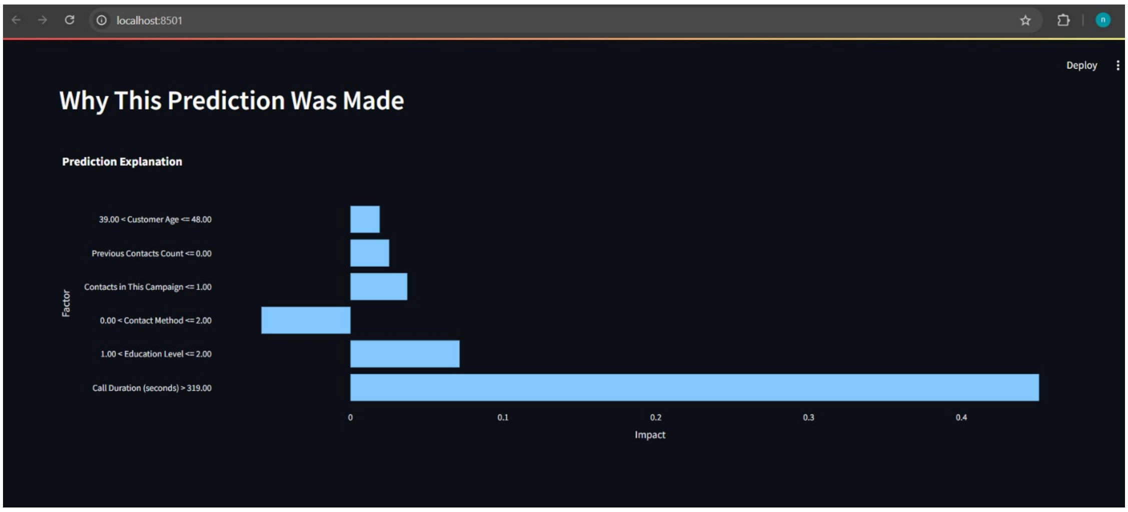
10. Output

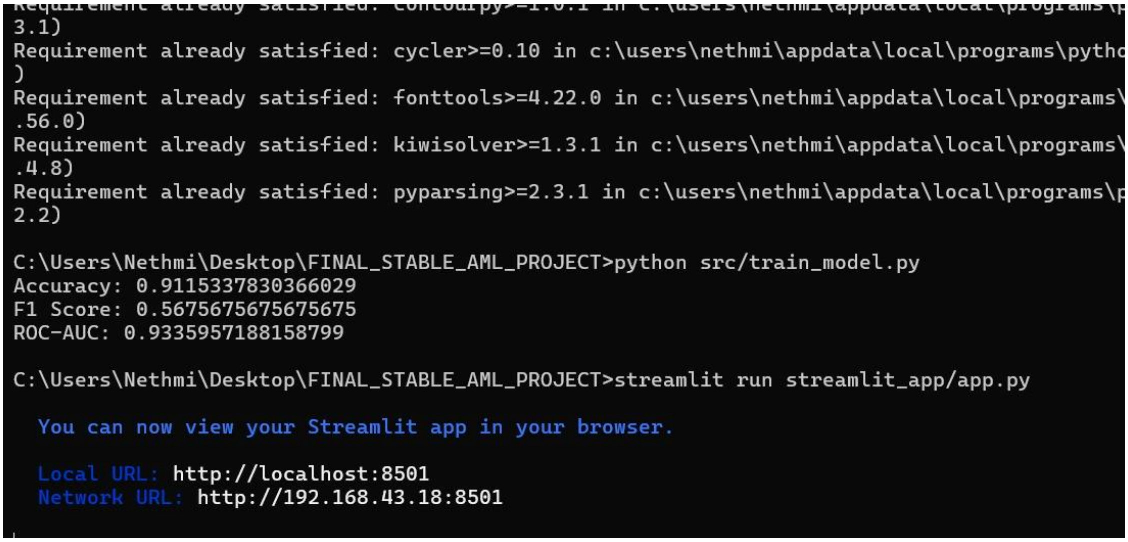

localhost:8501
Deploy
Bank Term Deposit Prediction
Simple, explainable prediction system
Customer Age
42.00
Occupation
management
Marital Status
married
Education Level
tertiary
Has Credit Default?
no
Account Balance
80.00


Has Personal Loan?
no
Contact Method
unknown
Last Contact Day
15.00
Last Contact Month
may
Call Duration (seconds)
850.00
Contacts in This Campaign
1.00
Days Since Last Contact
-1.00
Previous Contacts Count
0.00


Deploy
Last Contact Month
may
Call Duration (seconds)
850.00
Contacts in This Campaign
1.00
Days Since Last Contact
-1.00
Previous Contacts Count
0.00
Previous Campaign Outcome
unknown
Prediction Result
Unlikely to Subscribe
Probability: 0.33







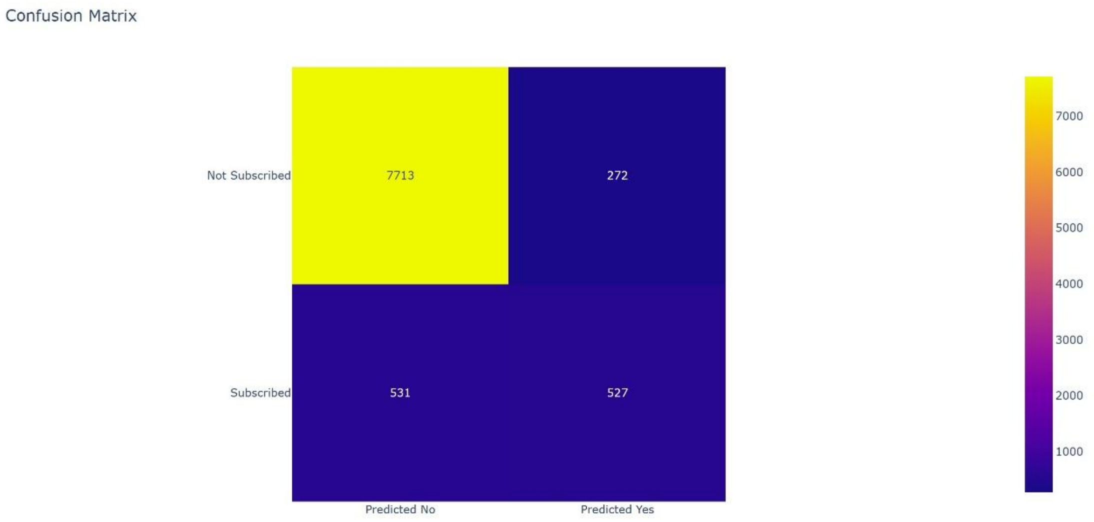
```
3.1)
Requirement already satisfied: cycler>=0.10 in c:\users\nethmi\appdata\local\programs\pytho
)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\nethmi\appdata\local\programs\
.56.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\nethmi\appdata\local\programs\
.4.8)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\nethmi\appdata\local\programs\p
2.2)

C:\Users\Nethmi\Desktop\FINAL_STABLE_AML_PROJECT>python src/train_model.py
Accuracy: 0.9115337830366029
F1 Score: 0.5675675675675675
ROC-AUC: 0.9335957188158799

C:\Users\Nethmi\Desktop\FINAL_STABLE_AML_PROJECT>streamlit run streamlit_app/app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.43.18:8501
```



11. Links

Dataset Link: https://archive.ics.uci.edu/dataset/222/bank+marketing

GithubLink: https://github.com/Suthan99/258826B_Assignment.git